

MUST lecture 4

Two-sample Hotelling T-squared test **Comparing mean vectors for two MVN samples**

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Two-sample **PAIRED** data test (statistically most powerful test)

Requires that each test object can be split in two parts and given "treatments" 1 & 2

Paired data model

$$(X_{1j}, X_{2j}), \quad j = 1, \dots, n$$
$$X_{1j} \sim N_p(\mu_1, \Sigma_1)$$
$$X_{2j} \sim N_p(\mu_2, \Sigma_2)$$

Difference model

$$D_j = X_{1j} - X_{2j} \sim N_p(\mu_1 - \mu_2, \Sigma_D)$$

Hypothesis

$$H_0: \mu_1 - \mu_2 = \delta_0 \quad \text{against} \quad H_1: \mu_1 - \mu_2 \neq \delta_0$$

Estimators

$$\hat{\mu}_1 - \hat{\mu}_2 = \bar{D} \sim N_p(\mu_1 - \mu_2, \Sigma_D/n)$$
$$\hat{\Sigma}_D = S_D = \frac{1}{n-1} \sum_{j=1}^n (D_j - \bar{D})(D_j - \bar{D})^T$$

Test statistic

$$T^2 \stackrel{\text{def}}{=} (\bar{D} - (\mu_1 - \mu_2))^T \left(\frac{S_D}{n} \right)^{-1} (\bar{D} - (\mu_1 - \mu_2)) \sim \frac{p(n-1)}{n-p} F(p, n-p)$$

Test

Put null-hypothesis in test statistic

$$\text{reject } H_0 \Leftrightarrow t^2 > \frac{p(n-1)}{n-p} F(p, n-p)_\alpha$$

Two-sample **NON-PAIRED** data test (statistically less powerfull test)

Is always possible (independent samples from two populations)

Non-paired data model $X_{1j} \sim N_p(\mu_1, \Sigma_1), \quad j = 1, \dots, n_1$
 $X_{2j} \sim N_p(\mu_2, \Sigma_2), \quad j = 1, \dots, n_2$

Hypothesis $H_0: \mu_1 - \mu_2 = \delta_0$ against $H_1: \mu_1 - \mu_2 \neq \delta_0$

Estimators $\hat{\mu}_1 = \bar{X}_1 \sim N_p(\mu_1, \Sigma_1/n_1)$
 $\hat{\mu}_2 = \bar{X}_2 \sim N_p(\mu_2, \Sigma_2/n_2)$

Case 1

$\Sigma_1 = \Sigma_2$ (homoscedasticity, tested with Bartlett test)

Pooled estimate

$$\hat{\Sigma} = S_P = \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1+n_2-2}$$

Test statistic $T^2 \stackrel{\text{def}}{=} ((\bar{X}_1 - \bar{X}_2) - \delta_0)^T \left(\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_P \right)^{-1} ((\bar{X}_1 - \bar{X}_2) - \delta_0) \sim \frac{p(n_1+n_2-2)}{n_1+n_2-p-1} F(p, n_1 + n_2 - p - 1)$

Test $\text{reject } H_0 \Leftrightarrow t^2 > \frac{p(n_1+n_2-2)}{n_1+n_2-p-1} F(p, n_1 + n_2 - p - 1)_\alpha$

Case 2

$\Sigma_1 \neq \Sigma_2$ (heteroscedasticity, tested with Bartlett test)

Large sample approximate test for $n_1 - p \gg 1, \quad n_2 - p \gg 1$

Test statistic $T^2 \stackrel{\text{def}}{=} ((\bar{X}_1 - \bar{X}_2) - \delta_0)^T \left(\frac{S_1}{n_1} + \frac{S_2}{n_2} \right)^{-1} ((\bar{X}_1 - \bar{X}_2) - \delta_0) \sim \chi^2(p)$

Test $\text{reject } H_0 \Leftrightarrow t^2 > \chi^2(p)_\alpha$

Bartlett test for homoscedasticity (is case 1 or 2 relevant ?)

Model

$$\begin{aligned} X_{1j} &\sim N_p(\mu_1, \Sigma_1), & j = 1, \dots, n_1 \\ X_{2j} &\sim N_p(\mu_2, \Sigma_2), & j = 1, \dots, n_2 \end{aligned}$$

Hypothesis

$$H_0: \Sigma_1 = \Sigma_2 \quad \text{against} \quad H_1: \Sigma_1 \neq \Sigma_2$$

Test statistic

$$T = c \cdot \{(n_1 + n_2 - 2) \log |S_P| - (n_1 - 1) \log |S_1| - (n_2 - 1) \log |S_2|\}$$

where the finite population correction factor

$$c = 1 - \frac{2p^2 + 3p - 1}{6(p+1)} \left(\frac{1}{n_1 - 1} + \frac{1}{n_2 - 1} - \frac{1}{n_1 + n_2 - 2} \right) \cong 1 \quad (\text{usually})$$

Approximately

$$T \sim \chi^2\left(\frac{1}{2}p(p+1)\right)$$

Test

$$\text{reject } H_0 \Leftrightarrow T > \chi^2\left(\frac{1}{2}p(p+1)\right)_\alpha$$